

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf



Part 4: Configure OSPF Passive Interfaces

a.

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0



What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129



Does R2 show up as an OSPF neighbor on R1? Yes



Does R2 show up as an OSPF neighbor on R3? No



What does this information tell you?

R1 can learn routes to the `192.168.2.0/24` network through R2 because R2's S0/0/0 interface is no longer passive, and an OSPF neighbor relationship has been established between R1 and R2.



R3 still learns routes to the `192.168.2.0/24` network through R1 because an OSPF neighbor relationship has not been established between R2 and R3, and R2's S0/0/1 interface remains passive.

This means that the `passive-interface default` command sets all interfaces to passive by default, but the `no passive-interface` command can be used to re-enable OSPF on specific interfaces.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1

no passive-interface s0/0/1



- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? [Serial0/0/1](#)

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

[65. This is the cost of R2 to 192.168.2.0/24 \(1\) plus the cost of R3 to R2 \(64\).](#)

Is R2 listed as an OSPF neighbor to R3? [Yes](#)

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

[Differentiating high-speed links, improves routing accuracy, provides more granular cost control, and optimizes routing based on specific needs.](#)

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

[Cost from R1 to 192.168.3.0/24:](#)

[Cost from R1 to R3 through Serial 0/0/1 = Reference Bandwidth\(100000k\) / Interface Bandwidth\(128k\)=781.](#)

[Cost for R3's g0/0:1](#)

[Total cost=781+1=782](#)

[Cost from R1 to 192.168.23.0/30:](#)

[Cost from R1 to R3 through Serial 0/0/1 = Reference Bandwidth\(100000k\) / Interface Bandwidth\(128k\)=781.](#)

[Cost for R3 to 192.168.23.0 through Serial 0/0/1:64](#)

[Total cost=781+64=845](#)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

[As the path is : R1-> R3 -> 192.168.23.0/30:](#)

Cost from R1 to R3 through Serial 0/0/1 = Reference Bandwidth(100000k) / Interface Bandwidth(128k)=781.

The cost of a connection from R3 to the 192.168.23.0/30 network through Serial0/0/1 = Reference Bandwidth(100000k) / Interface Bandwidth(128k) =781.

Total cost=781+781=1562.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The cost of the Serial0/0/1 interface from R1 to R3 is currently 1565. The cost of a direct connection from R3 to the 192.168.3.0/24 network is 1. Therefore, the total cost through R3 is 1565 + 1 = 1566.

The cost of the Serial0/0/0 interface from R1 to R2 is 781. The cost of the Serial0/0/1 interface from R2 to R3 is 781. The cost of a direct connection from R3 to the 192.168.3.0/24 network is 1. Therefore, the total cost through R2 is 781 + 781 + 1 = 1563.

OSPF always chooses the path with the lowest cost.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

A stable router ID prevents outages caused by unexpected changes such as interface jitter.

Router IDs are used in the DR/BDR election process in multi-access networks.

Controlling the router ID can influence which router becomes the DR.

A well-designed router ID scheme can simplify network troubleshooting; consistent router IDs can help prevent routing loops in complex scenarios.

2. Why is the DR/BDR election process not a concern in this lab?

This lab primarily uses point-to-point serial links. DR/BDR election is only relevant in multi-access networks.

3. Why would you want to set an OSPF interface to passive?

Passive interfaces do not send or receive OSPF hello packets, thus preventing adjacency formation; however, routers still advertise networks connected to the passive interface, allowing other routers to learn about that network; passive interfaces enhance security by preventing unauthorized routers from establishing adjacencies.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

In OSPF's area structure:

Areas: OSPF allows an AS to be divided into multiple areas.

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Area 0 (Backbone Area): All areas are connected to Area 0.

Area Border Routers (ABRs): ABRs connect to multiple areas (including Area 0).

Internal Routers: Routers within an area only have detailed information about their own area.

Summary: ABRs summarize routing information between areas.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

EIGRP with a metric of 4039043. Because it has the lowest Administrative Distance (AD) (EIGRP=90, OSPF=110, RIP=120).